

Digital Image Processing

Assignment 02: Image Enhancement

1. Objective

The objective of this experiment is to evaluate the results of applying several image enhancement methods to different image conditions. The methods used in this experiment are Laplacian Sharpening, Log Transform, Histogram Equalization, Gamma Transform, and Contrast Stretching. Each method is implemented according to the image condition that is suitable for the respective method.

2. Blur Image Enhancement Analysis

For the blurred image case, the method used in this experiment is Laplacian Sharpening. The method works by calculating the difference between the intensity of the target pixel and the intensities of its neighboring pixels. The differences are then summed to obtain the total change in intensity. This value is multiplied by an alpha parameter and added to the intensity of the target pixel to produce the sharpened pixel value. In this experiment, two Laplacian methods are implemented, namely 4-neighbor and 8-neighbor Laplacian. Both methods use the same calculation process, with the difference being the number of neighboring pixels considered.

Based on the experimental results, Laplacian Sharpening produces a sharper image by emphasizing the intensity differences around the edges. The value of alpha also affects the resulting sharpness. A higher alpha value produces a stronger sharpening effect because the Laplacian contribution added to the original pixel intensity becomes larger. However, an excessively high alpha value can cause the edges to appear overly emphasized. Overall, the results show that Laplacian Sharpening can be applied to improve the sharpness of a blurred image.

3. Dark Image Enhancement Analysis

For the dark image case, the methods used in this experiment are Log Transform and Histogram Equalization. These two methods are not directly related, but they have the same objective, which is to enhance the brightness and visibility of a dark image. The Log Transform method works by transforming the intensity of each pixel using the formula $s = c \log(1 + r)$, where c is the scaling constant and r is the original pixel intensity. Meanwhile, Histogram Equalization transforms the pixel intensities based on the overall intensity distribution of the image. Based on the results, the Log Transform produces an image that appears relatively pale and bright, while Histogram Equalization produces a brighter image in which the objects contained in the image appear clearer. This difference occurs because the two methods use different approaches to enhance the image, with Log Transform applying a logarithmic transformation to each pixel and Histogram Equalization adjusting the intensity values based on the image's intensity distribution.

4. Bright Image Enhancement Analysis

For the bright image case, the method used in this experiment is Gamma Transform. This method works by modifying the intensity of each pixel using the formula $s = 255((r/255)^\gamma)$, where r represents the original pixel intensity and γ is the gamma value. Based on the formula, a higher gamma value produces a lower resulting intensity for pixel values between 0 and 255. This behavior is consistent with the objective of the experiment, which is to reduce the excessive brightness of the image. Based on the experimental results, the effect of increasing the gamma value can be clearly observed, as the resulting image becomes progressively darker as the gamma value increases.

5. Low Contrast Image Enhancement

For the low contrast image case, the method used in this experiment is Contrast Stretching. This method transforms the intensity of each pixel using the formula $s = \frac{r - r_{min}}{r_{max} - r_{min}} \times 255$, where r represents the original pixel intensity, while r_{min} and r_{max} represent the minimum and maximum intensity values used for the transformation. In this experiment, three different intensity ranges are applied, namely 90–190, 100–180, and 110–170. Based on the experimental results, the 90–190 range produces an image with increased contrast while maintaining relatively smooth transitions. The 100–180 range produces more pronounced edges, making the lines in the image appear sharper. Meanwhile, the 110–170 range produces the strongest contrast, resulting in significantly sharper edges. These results show that using a narrower intensity range produces a stronger contrast enhancement effect.